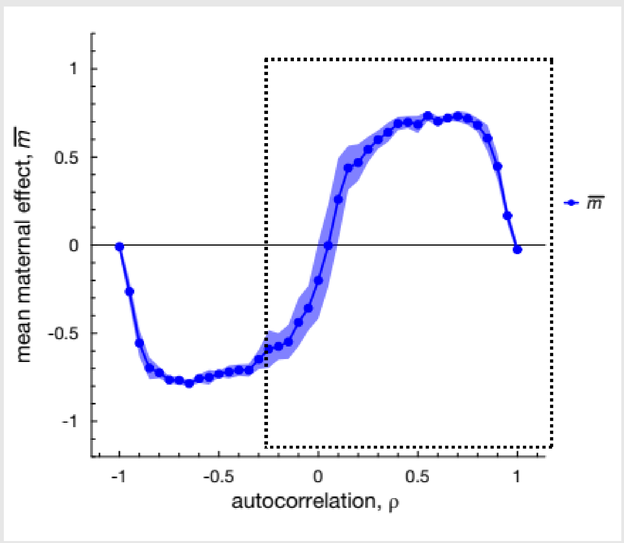
When Phenotypic Memory Hinders Adaptation: Parental Effects in Stochastically Fluctuating Environments



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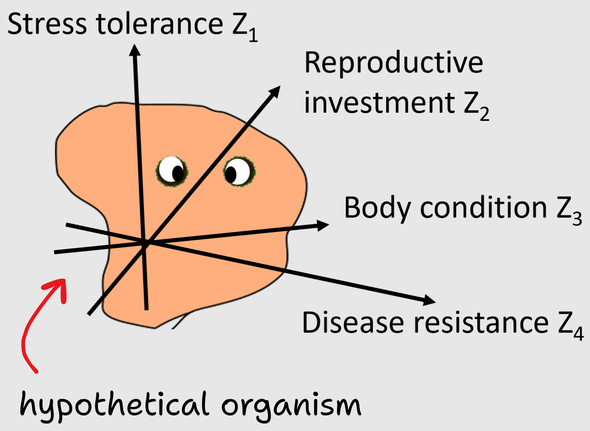
Published models show that Parental effects are adaptive!



Kuijper et al 2014[1]

Previosuly positive parental effects always evolves in autocorrelated, stable environments

But what if multiple traits contribute to fitness?



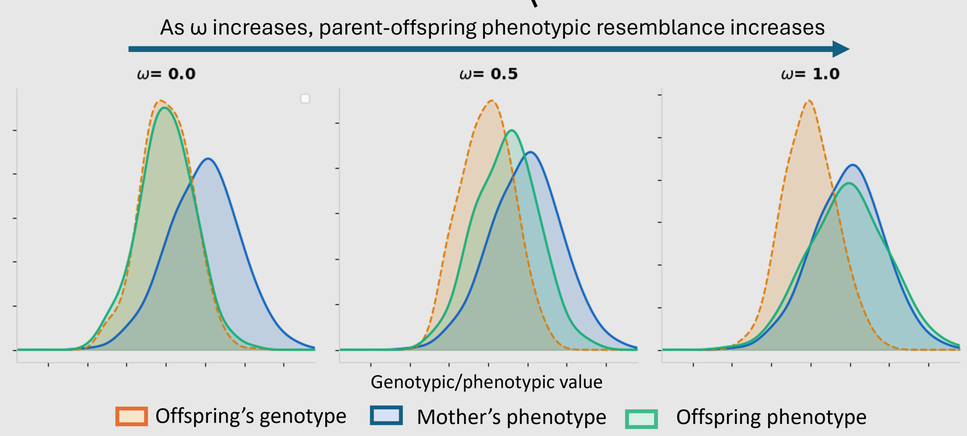
Older models assume only one trait contributes to fitness. But, we know from classical adaptation theory that when multiple traits contribute to fitness, adaptation becomes harder

Theoretical Model

Trait vector Z = (z1, z2, …., zn) ... multiple traits

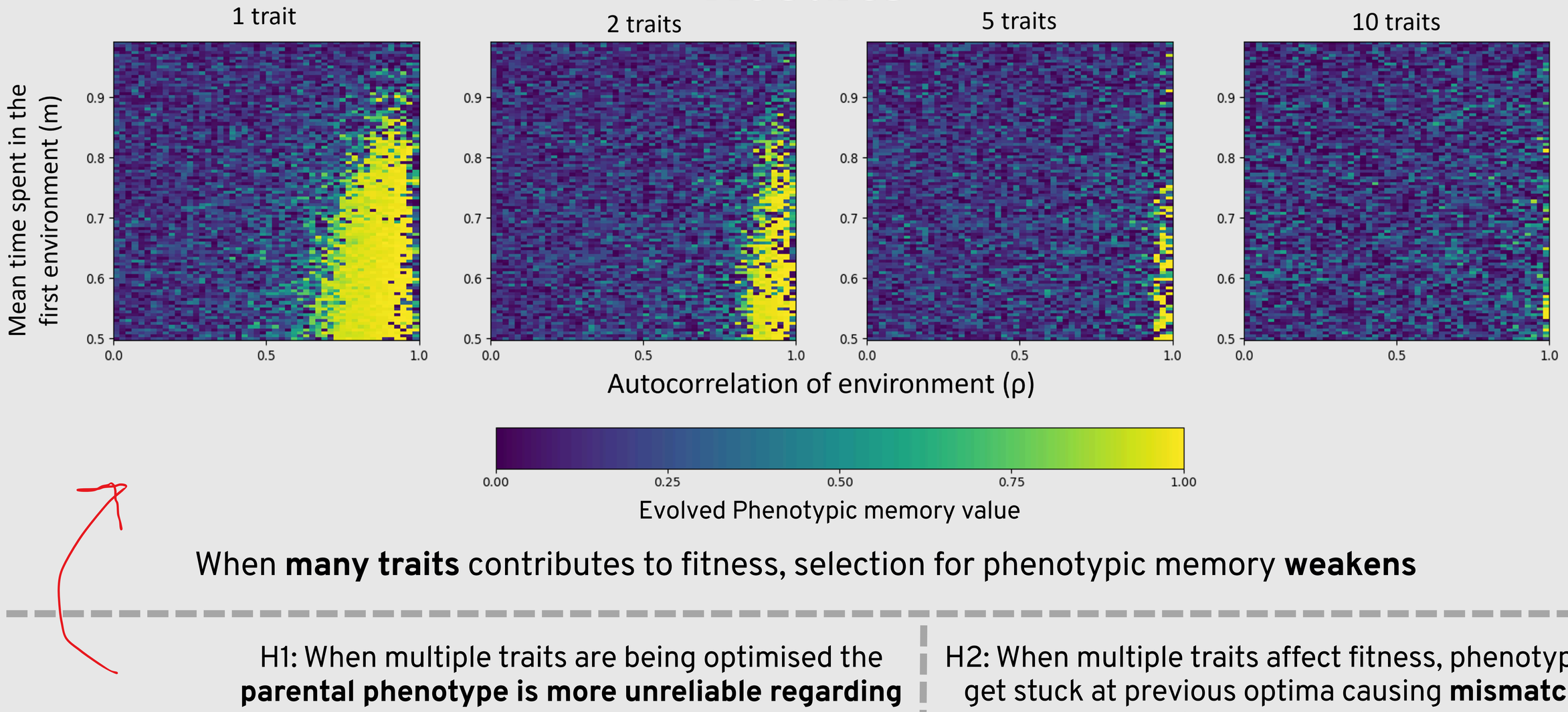
$$Z_{offspring} = G_{offspring} + \omega * (Z_{parent} - G_{parent}) + \varepsilon_{noise}$$

ω = ?



We consider an ***evolving phenotype Z*** with multiple traits contributing to fitness, and aim to understand when and to what strength **phenotypic memory (ω)** evolves, when adapting to **a two state fluctuating environment**

Results

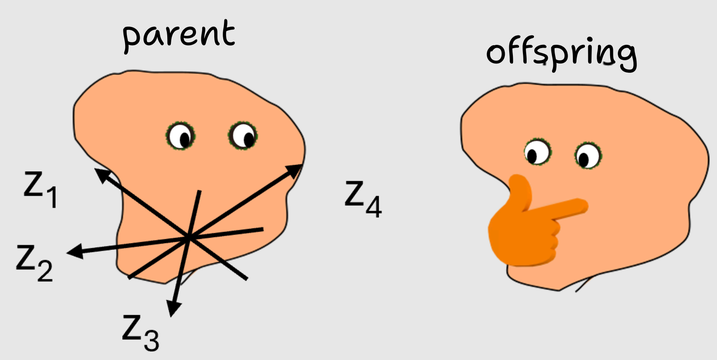


When **many traits** contributes to fitness, selection for phenotypic memory **weakens**

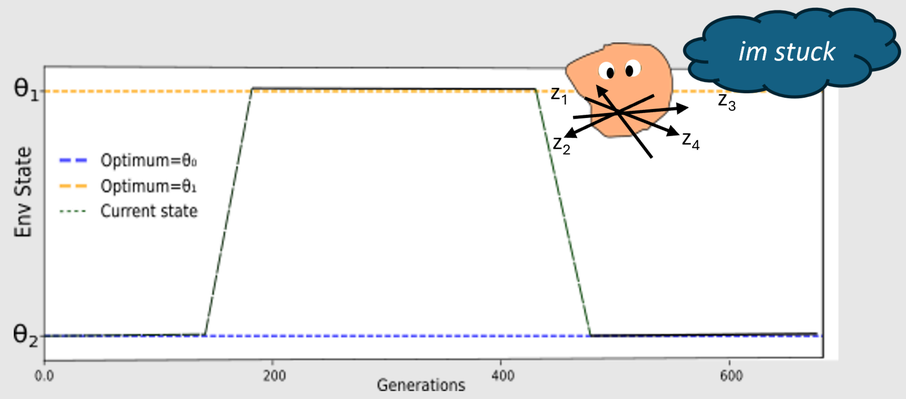
Why?

Two non-exclusive hypothesis:

H1: When multiple traits are being optimised the parental phenotype is more unreliable regarding current conditions



H2: When multiple traits affect fitness, phenotypes get stuck at previous optima causing mismatch costs to accumulate



Implications: Then what explains the evolution of known cases of phenotypic memory (e.g epigenetic inheiritence, maternal provisioning via yolk) ?

Have feedback? Reach out!
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1. Kuijper, B., Johnstone, R. A., & Townley, S. (2014). The evolution of multivariate maternal effects. PLoS Computational Biology, 10(4), e1003550.